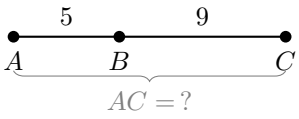
Segment Addition Postulate — Practice — Answer Key

A worked example comes before each type. In every problem, the middle point is between the other two unless it says otherwise. Show your thinking.

Name: _____ Date: _____

Type 1 Find the Whole: Add

Watch Me — Worked Example



B is between A and C, so part + part = whole.
$AC = AB + BC = 5 + 9$. **Answer: $AC = 14$**

1. $AB = 12$ and $BC = 15$. Find AC.

$AC = 27$

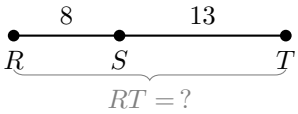
2. $AB = 3.5$ and $BC = 6.2$. Find AC.

$AC = 9.7$

3. $AB = 2\frac{1}{2}$ and $BC = 4\frac{1}{4}$. Find AC.

$AC = 6\frac{3}{4}$

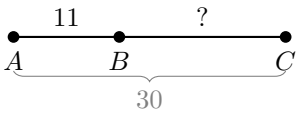
4. Find RT.



$RT = 21$

Type 2 Find a Missing Piece: Subtract

Watch Me — Worked Example



You know the whole and one piece, so subtract.
$BC = AC - AB = 30 - 11$. **Answer: $BC = 19$**

5. $AC = 25$ and $BC = 9$. Find AB.

$AB = 16$

6. $AC = 17.5$ and $AB = 6.8$. Find BC.

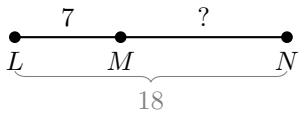
$BC = 10.7$

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7. Q is between P and R. $PR = 52$ and $QR = 29$. Find PQ.

$PQ = 23$

8. Find MN.



$MN = 11$

Type 3 Solve with Algebra

Watch Me — Worked Example

$AB = 2x + 1,\ BC = 3x - 4,\ AC = 27$. Find x, AB and BC.

$(2x + 1) + (3x - 4) = 27 \Rightarrow 5x - 3 = 27 \Rightarrow 5x = 30 \Rightarrow x = 6$.

Don't stop at x: $AB = 2(6) + 1 = 13,\ BC = 3(6) - 4 = 14$. Check: $13 + 14 = 27$ ✓

Answer: $x = 6,\ AB = 13,\ BC = 14$

9. $AB = x + 4,\ BC = 3x,\ AC = 24$. Find x and BC.

$x = 5,\ BC = 15$

10. K is between J and L. $JK = 2x - 3,\ KL = x + 6,\ JL = 18$. Find x and JK.

$x = 5,\ JK = 7$

11. E is between D and F. $DE = 4x - 5,\ EF = 2x + 1,\ DF = 32$. Find x and EF.

$x = 6,\ EF = 13$

12. M is between A and B. $AM = 3x + 2,\ MB = 5x - 6,\ AB = 36$. Find x and AM.

$x = 5,\ AM = 17$

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Type 4 Is It Between? & Number Lines

Watch Me — Worked Example

$AB = 4,\ BC = 6,\ AC = 9$. Is B between A and C?
If it were, then $AB + BC$ would equal AC. But $4 + 6 = 10 \neq 9$. **Answer: no.**
Number line: the distance between two points is the difference of their coordinates, like $|9 - 2| = 7$.

13. $AB = 7,\ BC = 5,\ AC = 12$. Is B between A and C? Why?

yes: $7 + 5 = 12$

14. $AB = 8,\ BC = 3,\ AC = 10$. Is B between A and C? Why?

no: $8 + 3 = 11 \neq 10$

15. On a number line, A is at -3, B is at 2 and C is at 9. Find AB, BC and AC.

$5,\ 7,\ 12$

16. HARD B is between A and C, $AB : BC = 2 : 3$, and $AC = 30$. Find AB.

$AB = 12$

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